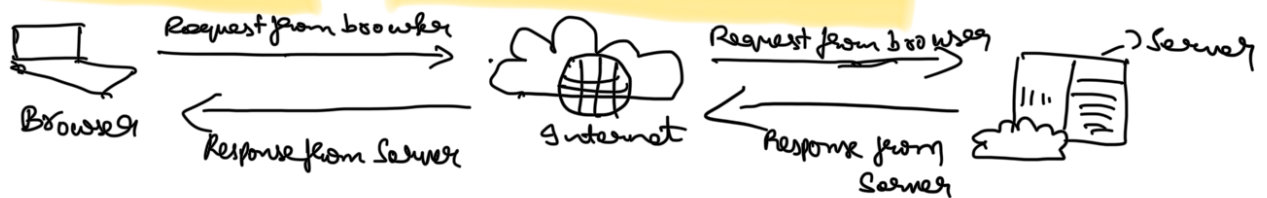


How websites work :-

When you visit website, your browser (like Safari or Google Chrome) makes a request to web server asking for information about page you're visiting. It'll respond with data that your browser uses to show your page; a web server is just dedicated computer somewhere else in world that handles your request.



There are 2 major components that make up website :-

1. Front End (Client-Side) :- The way your browser renders a website.
 to take code (HTML, JS, CSS) and translate to visual representation.
2. Back End (Server-Side) :- A server that processes your request and returns a response.

Quiz :-

1) What term best describes the component of web application rendered by your browser?

A) Front End

Task 2 :- HTML

Websites are primarily created using :-

- 1) HTML, to build websites and define their structure
- 2) CSS, to make websites look pretty by adding styling options
- 3) Javascript implement complex features on pages using interactivity.

Hypertext Markup language (HTML) is language websites are written in. Elements (also known as tags) are building blocks of HTML pages and tells browser how to display content. Let's see structure of HTML

```
<!DOCTYPE html>
<html>
  <head>
    <title> Page Title </title>
  </head>
```

```

</body>
  <h1> Example heading </h1>
  <p> Example paragraph </p>
</body>

```

</html>

The HTML structure (as shown above) has following components

* The <!DOCTYPE HTML> defines that page is HTML5 document. This helps standardisation across different browsers and tells browser to use HTML5 to interpret the page.

* The <html> element is root element of HTML page - all other elements come after this element.

* The <head> element contains information about page (such as page title)

* The <body> element defines the HTML documents body. Only content inside of body shown in browser.

* The <h1> defines a large heading

* The <p> element defines a paragraph.

* There are many other elements used in different purposes. Ex: - There are tags for

buttons (<button>), image (), lists and much more

* Tags contain attributes such as class attribute which can be used to style an element.

<p class="bold-text">, or src attribute which is used in image to specify location of image.

. An element can have multiple attributes each with its own unique purpose.

Ex: - <p attribute 1="value 1" attribute 2="value 2">

* Elements can also have an id attribute (<p id="example">). which is unique to element.

~~you can~~

→ You can view HTML of any website by right clicking and selecting "View page source" (Chrome) /

"Show page source" (Safari).

Quiz: -

Q) One of the images on cat website is broken - fix it and guess will reveal hidden tent answer!

A) HTML HERO
HTML code

<!DOCTYPE HTML>

Q) Add a dog image to page by adding another img tag on line 11. The dog image location is img/dog1.png. What is tent in dog image?

A) DOG HTML

```

<html>
  <head>
    <title> TryHackMe HTML Editor </title>
  </head>
  <body>
    <h1> Cat website </h1>
    <p> see images of all my cats! </p>
    <img src = 'img /cat1.jpg'>
    <img src = 'img /cat2.jpg'>
    <!-- Add dog image here -->
    <img src = 'img /dog-1.png'>
  </body>
</html>

```

code
Reveals
Quiz
answers

Task 3 :- Javascript

JS (Javascript) is most popular coding languages in world allows pages to become interactive. HTML is used to create website structure and content, while JS is used to control functionality of web pages. without javascript a page would not have interactive elements would always be static. JS is dynamically updated page in real time, giving functionality to change style of buttons when particular event on page occurs or to display moving animation.

* Javascript is added within page source code and can be either loaded within

<script> tags or can be included remotely with the src attribute <script src = "/location/of/javascript_file.js"></script>

* The following js code finds HTML element of page with id of "demo" and changes element's content to "Hack the planet": `document.getElementById("demo").innerHTML = "Hack the planet";`

* HTML elements can also have events, such as "onclick" or "onhover" that execute Javascript when event occurs. The following code changes the text of element with demo ID to button clicked: `<button onclick = 'document.getElementById("demo").innerHTML = "Button clicked";'> Click Me! </button>` - on click events can also be defined inside JS tags, not elements directly.

HTML + JS code

```
<!DOCTYPE HTML>
```

```
<html>
```

```
  <head>
```

```
    <title> Try Hack Me editor </title>
```

```
  </head>
```

```
  <body>
```

```
    <div id = "demo" > Hi there! </div>
```

```
    <script type = "text/javascript">
```

```
      // add you JS here
```

```
      document.getElementById("demo").innerHTML = "Hack the planet";
```

```
    </script>
```

```
    <button onclick = 'document.getElementById("demo").innerHTML = "button
```

```
clicked";' > Click Me! </button>
```

```
  </body>
```

```
</html>
```

Output:-

Hack the planet

Click Me!